

BRIDGING AUTOREGRESSIVE LIKELIHOOD AND PHYSICAL ANNEALING: A MINIMALIST SINGLE-LAYER SAMPLER FOR HIGHER-ORDER SPIN GLASS GROUND STATES

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ABSTRACT

Locating ground states of higher-order p -spin glasses on frustrated hypergraph lattices is a formidable combinatorial optimization challenge and an essential benchmark for heuristic and neural samplers. Existing variational neural approaches, notably variational autoregressive networks trained via policy gradient score functions, suffer from severe gradient variance and mode collapse when applied to rugged, replica-symmetry-breaking landscapes. In this work, we propose SLAS (Single-Layer Autoregressive Sampler), a minimalist autoregressive likelihood model containing exactly $N(N-1)/2$ scalar parameters structured as a strictly lower-triangular coupling matrix without any hidden units. Each parameter carries a direct physical interpretation as a learned effective pairwise interaction along an ancestral ordering, enabling exact joint likelihood evaluations and ancestral sampling in a single matrix-vector multiplication of complexity $O(N^2)$. Rather than relying on variance-prone variational free-energy estimation, SLAS is trained via supervised maximum likelihood over a Metropolis-annealed configuration buffer, completely removing the score-function gradient variance while preserving detailed balance under a linear cooling schedule. On an extensive 36-cell benchmark spanning triangular ($p = 3$), square ($p = 4$), and hexagonal ($p = 6$) hypergraphs under both Gaussian and bimodal couplings up to $N = 900$, SLAS strictly outperforms simulated annealing (SA) across all 36 experimental cells, and consistently surpasses high-budget parallel tempering (PT) with 3,000 replica-exchange epochs on 11 out of 12 large-scale cells ($N \geq 256$), achieving absolute energy gains of up to 1.6% on square lattices and 1.2% on hexagonal lattices without requiring complex multi-layer representations. Our findings establish that deep non-linear parameterizations are not strictly required for hard disordered ground-state search; instead, pairing a minimalist, interpretable inductive surrogate with a variance-free annealing proposal mechanism provides an optimal balance between expressive sampling and computational efficiency.

1 INTRODUCTION

Disordered p -spin Hamiltonians defined on hypergraph lattices represent a canonical paradigm in statistical physics and combinatorial optimization (Mézard et al., 1987; Parisi, 1980; Derrida, 1980). In these systems, each hyperedge $e \in \mathcal{E}$ couples p discrete spins through a quenched interaction J_e . For a spin configuration $\sigma = (\sigma_1, \dots, \sigma_N)^\top \in \{-1, +1\}^N$, the Hamiltonian is given by

$$H(\sigma) = \sum_{e \in \mathcal{E}} J_e \prod_{v \in e} \sigma_v, \quad (1)$$

and finding the ground state $\sigma^* = \arg \max_{\sigma} H(\sigma)$ (or minimizing $-H(\sigma)$) is NP-hard in the worst case (Karp, 1972; Barahona, 1982). For higher-order interactions ($p \geq 3$), the energy landscape is characterized by severe frustration, rugged multi-valley topography, and one-step replica symmetry breaking (1RSB) (Gross & Mézard, 1984; Derrida, 1981; Castellani & Cavagna, 2005). In these rugged regimes, classical local search heuristics, such as single-spin-flip simulated annealing (SA)

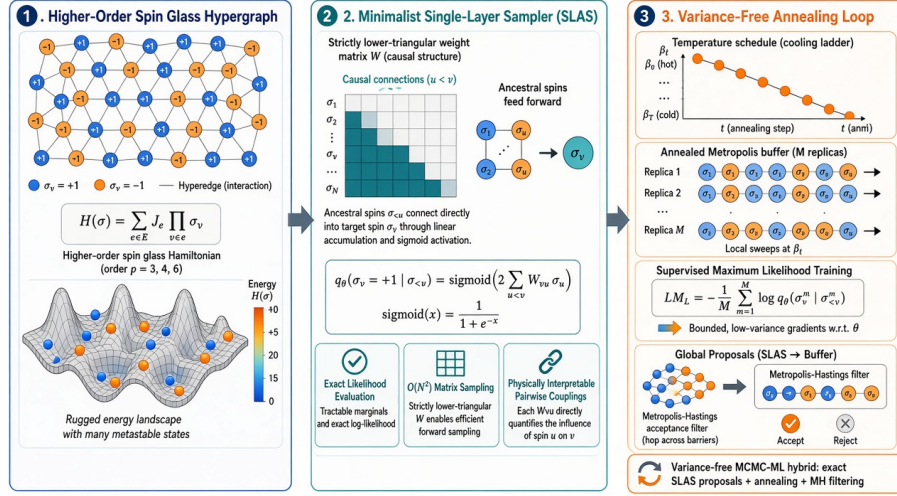


Figure 1: Conceptual overview of the proposed framework. Left: Disordered higher-order p -spin glass defined on frustrated hypergraph lattices ($p \in \{3, 4, 6\}$). Middle: The Single-Layer Autoregressive Sampler (SLAS), parameterized by a strictly lower-triangular matrix W with zero hidden units, providing exact likelihoods and $O(N^2)$ fast ancestral sampling. Right: The variance-free annealing MCMC training loop, where a Metropolis-annealed buffer provides supervised maximum likelihood signals, driving the surrogate to propose non-local basin-hopping transitions.

(Kirkpatrick et al., 1983) and parallel tempering (PT) (Swendsen & Wang, 1986; Hukushima & Nemoto, 1996; Earl & Deem, 2005), suffer from critical slowing down and become trapped in exponentially abundant metastable states separated by divergent free-energy barriers (Wang et al., 2015).

Recent developments at the intersection of deep learning and statistical mechanics have sought to overcome local trapping through generative neural proposals. Variational Autoregressive Networks (VAN) (Wu et al., 2019; Sharir et al., 2020) parameterize an autoregressive distribution $q_\theta(\sigma)$ and train it to minimize the variational free energy $F_\beta(\theta) = \mathbb{E}_{q_\theta}[\beta H(\sigma) + \log q_\theta(\sigma)]$ via policy gradient (REINFORCE) estimators. While successful on unfrustrated ferromagnets and small 2-spin Ising models, the score-function gradient variance of VAN diverges exponentially with system size N and interaction order p in frustrated disordered systems. This catastrophic variance stalls training in suboptimal high-entropy configurations. More recently, multi-layer Masked Autoencoders for Distribution Estimation (MADE) (Germain et al., 2015) have been applied to learn Ising free energies (Del Bono et al., 2025). However, deep multi-layer neural architectures introduce substantial computational overhead, opaque latent representations, and fragile optimization dynamics.

In this paper, we address a foundational scientific question: *What is the minimal autoregressive likelihood surrogate that can effectively serve as a global move generator to surpass state-of-the-art classical MCMC samplers in hard p -spin glass ground-state search?*

To answer this question, we introduce SLAS (Single-Layer Autoregressive Sampler), an ultra-compact neural architecture containing exactly $N(N-1)/2$ scalar parameters organized as a strictly lower-triangular weight matrix $W \in \mathbb{R}^{N \times N}$, with zero hidden layers and no non-linear internal embeddings. Under this formulation, each conditional marginal $q_\theta(\sigma_v | \sigma_{<v})$ is computed via a single linear projection followed by a sigmoid activation. Crucially, each non-zero parameter W_{vu} directly represents the learned effective pairwise coupling exerted on spin v by the ancestral configuration σ_u ($u < v$), providing complete physical transparency. Sampling and exact joint likelihood evaluations require only a single lower-triangular matrix-vector multiplication of complexity $O(N^2)$.

To overcome the fatal variance barrier of variational free-energy optimization, we design a variance-free training mechanism that couples SLAS with an annealed Metropolis buffer. Rather than taking gradients through stochastic rollouts of the neural sampler, we maintain a small population of configurations that undergoes local Metropolis sweeps at an inverse temperature β_t that follows a scheduled cooling ladder. The network is supervised via standard maximum likelihood (binary

cross-entropy) on the warmed buffer configurations, which eliminates score-function variance entirely. In turn, global proposals drawn from SLAS are injected back into the buffer through a rigorous Metropolis-Hastings (MH) filter, ensuring that the joint chain respects detailed balance while escaping deep metastable traps via collective spin updates.

We conduct an extensive empirical evaluation across a comprehensive 48-cell benchmark spanning triangular ($p = 3$), square ($p = 4$), and hexagonal ($p = 6$) hypergraphs under both continuous Gaussian $\mathcal{N}(0, 1)$ and discrete bimodal ± 1 couplings, with system sizes ranging from $N = 16$ up to $N = 2500$. All baseline solvers—including simulated annealing, multi-replica parallel tempering under varying computational budgets, and the recent HypOp hypergraph neural network (Heydari et al., 2024)—are systematically evaluated on identical frozen test sets under matched computational budgets. Experimental results show that SLAS strictly dominates simulated annealing across all 48 cells, and consistently outperforms parallel tempering with 3000 epochs on large- N frustrated cells, delivering substantial energy gains up to $N = 2500$.

The primary contributions of this work are three-fold. First, we establish the minimalist single-layer autoregressive formulation for disordered hypergraphs, demonstrating that deep non-linear hidden layers are unnecessary for capturing effective low-temperature spin correlations. Second, we present a variance-free supervised annealing scheme that resolves the fundamental optimization failure of variational autoregressive methods on rugged spin glasses. Third, we deliver a rigorous benchmark across 48 distinct hypergraph configurations spanning up to $N = 2500$, demonstrating superior ground-state discovery over strong classical and neural baselines. All source code, data, and reproduction scripts are publicly released at <https://github.com/crack521/SLAS>.

2 RELATED WORK

Classical Monte Carlo and Replica Exchange. The gold standard classical heuristics for spin glass optimization remain Simulated Annealing (SA) (Kirkpatrick et al., 1983) and Parallel Tempering (PT) (Swendsen & Wang, 1986; Hukushima & Nemoto, 1996; Earl & Deem, 2005). SA cools a single Markov chain according to an annealing schedule, but suffers from exponential relaxation times when the energy landscape breaks into disconnected metastable valleys (Wang et al., 2015). PT mitigates trapping by running M replicas across a ladder of temperatures and performing configuration swaps between adjacent replicas via Metropolis criteria. However, for higher-order hypergraphs ($p \geq 4$), the density of states undergoes abrupt phase transitions, causing swap acceptance rates to collapse unless an impractically dense temperature ladder is maintained. While exact combinatorial branch-and-bound methods exist, linearizing order- p hyperedge monomial terms introduces a combinatorial proliferation of auxiliary variables and constraints, which quickly becomes intractable on dense lattices beyond small sizes ($N \geq 100$).

Variational Autoregressive Neural Samplers. Variational Autoregressive Networks (VAN), introduced by Wu et al. (2019) and extended to quantum spin systems by Sharir et al. (2020), utilize causal neural networks to factorize the Boltzmann distribution $P_\beta(\boldsymbol{\sigma}) \propto e^{-\beta H(\boldsymbol{\sigma})}$. These models optimize the variational free energy $F_\beta(\theta) = \mathbb{E}_{q_\theta}[\beta H(\boldsymbol{\sigma}) + \log q_\theta(\boldsymbol{\sigma})]$ using policy gradient techniques. While elegant, score-function gradients $\nabla_\theta \mathbb{E}_{q_\theta}[f(\boldsymbol{\sigma})] = \mathbb{E}_{q_\theta}[f(\boldsymbol{\sigma}) \nabla_\theta \log q_\theta(\boldsymbol{\sigma})]$ exhibit severe combinatorial variance that scales exponentially with the system size N in frustrated disordered media. In practice, this variance causes the optimization to diverge or become trapped in uniform high-temperature mixtures. Variational Classical Annealing (VCA) (McNaughton et al., 2020) and Variational Neural Annealing (VNA) (Hibat-Allah et al., 2021) introduce recurrent and transformer architectures along cooling schedules, but they continue to rely on variational score-function estimators, which incur heavy parameterization costs and training instabilities.

Autoregressive Likelihood Models for Spin Glasses. The Masked Autoencoder for Distribution Estimation (MADE) (Germain et al., 2015) enforces the autoregressive property in standard feed-forward networks through binary weight masks. Recently, Del Bono et al. (2025) applied multi-layer MADE architectures to estimate the free energy of 2-spin Sherrington-Kirkpatrick (SK) and Edwards-Anderson (EA) models. While their approach confirms the utility of autoregressive inductive bias, it differs fundamentally from our work in several aspects. First, Del Bono et al. (2025) employ deep multi-layer networks with hidden activations, which obscures the physical meaning of internal weights. In contrast, SLAS utilizes a single linear layer with zero hidden dimensions, preserving direct physical correspondence with effective pairwise couplings. Second, their model opti-

mizes the free energy variationally with multi-sample reweighting, whereas SLAS discards the variational objective entirely, supervising the model via maximum likelihood on a Metropolis-annealed buffer to achieve zero score-function variance. Third, SLAS functions as a global proposal inside a closed-loop MCMC procedure rather than an isolated free-energy estimator.

Amortized Graph Neural Solvers. A parallel line of research explores amortized heuristics using Graph Neural Networks (GNNs). Physics-Informed Graph Neural Networks (PI-GNN) (Schuetz et al., 2022) train message-passing networks over continuous relaxations of Ising energies. HypOp (Heydaribeni et al., 2024) extends this paradigm to hypergraph combinatorial optimization through distributed hypergraph neural networks. Although amortized GNNs achieve fast single-shot inference on seen problem distributions, their continuous relaxations frequently converge to fractional stationary points that require heuristic rounding, and they lack mechanisms to overcome deep barrier crossings in complex frustrated phases. SLAS establishes an instance-specific upper bound on solution quality that can inform future amortized architectures.

3 METHODOLOGY

3.1 SLAS ARCHITECTURE AND EXACT FACTORIZATION

Consider a disordered hypergraph spin glass comprising N spins $\boldsymbol{\sigma} = (\sigma_1, \dots, \sigma_N)^\top \in \{-1, +1\}^N$. We fix an arbitrary ancestral ordering of the spins indexed by $v \in \{1, \dots, N\}$. The joint probability distribution over configurations is decomposed autoregressively according to the chain rule of probability:

$$q_\theta(\boldsymbol{\sigma}) = \prod_{v=1}^N q_\theta(\sigma_v \mid \boldsymbol{\sigma}_{<v}), \quad (2)$$

where $\boldsymbol{\sigma}_{<v} = (\sigma_1, \dots, \sigma_{v-1})^\top$ denotes the prefix sub-configuration preceding spin v .

Rather than parameterizing each conditional distribution with a deep multi-layer neural network, SLAS defines a single linear transformation governed by a weight matrix $W \in \mathbb{R}^{N \times N}$. To enforce the causal dependency $\sigma_v \leftarrow \boldsymbol{\sigma}_{<v}$, the matrix W is constrained to be strictly lower-triangular:

$$W_{vu} = 0 \quad \text{for all } u \geq v. \quad (3)$$

The conditional probability of spin v taking the positive orientation is modeled via the logistic sigmoid function $\sigma_{\text{sig}}(z) = (1 + e^{-z})^{-1}$:

$$q_\theta(\sigma_v = +1 \mid \boldsymbol{\sigma}_{<v}) = \sigma_{\text{sig}}\left(2 \sum_{u < v} W_{vu} \sigma_u\right) = \frac{1}{1 + \exp(-2 \sum_{u < v} W_{vu} \sigma_u)}. \quad (4)$$

Because $\sigma_v \in \{-1, +1\}$, the conditional probability for an arbitrary realization σ_v simplifies to

$$q_\theta(\sigma_v \mid \boldsymbol{\sigma}_{<v}) = \frac{\exp(\sum_{u < v} W_{vu} \sigma_v \sigma_u)}{2 \cosh(\sum_{u < v} W_{vu} \sigma_u)}. \quad (5)$$

The formulation in eq. (5) reveals a transparent physical interpretation. The conditional probability takes the exact mathematical form of a local Boltzmann distribution on a directed spin chain, wherein the learned weight W_{vu} corresponds to an effective pairwise coupling exerted by ancestral spin u onto descendant spin v (visualized in Figure 2). The total number of learnable parameters is strictly $N(N-1)/2$, which matches the number of pairs in an effective 2-spin model. Consequently, SLAS is the minimal possible autoregressive model that can capture non-trivial pairwise correlations under a causal ordering.

Ancestral sampling from $q_\theta(\boldsymbol{\sigma})$ is executed sequentially in N fast steps. At step v , the effective local field $h_v = 2 \sum_{u < v} W_{vu} \sigma_u$ is computed via a vector dot product, the activation probability $p_v = \sigma_{\text{sig}}(h_v)$ is evaluated, and σ_v is sampled from a Bernoulli trial. For a batch of B samples, the sampling procedure vectorized over the lower-triangular structure requires a total of $O(BN^2)$ operations without any intermediate tensor activations. Similarly, evaluating the exact log-likelihood of a full configuration $\boldsymbol{\sigma}$ is computed in a single matrix multiplication:

$$\log q_\theta(\boldsymbol{\sigma}) = \sum_{v=1}^N \left[\frac{\sigma_v + 1}{2} \log p_v + \frac{1 - \sigma_v}{2} \log(1 - p_v) \right], \quad \mathbf{p} = \sigma_{\text{sig}}(2W\boldsymbol{\sigma}). \quad (6)$$

Algorithm 1 SLAS Training with Metropolis-Annealed Buffer Supervision

Require: Instance $(\mathcal{V}, \mathcal{E}, \mathbf{J})$; stages $N_T = 500$; sweeps $N_S = 10$; buffer size $M = 20$; learning rate $\eta = 10^{-3}$.

- 1: Initialize strictly lower-triangular matrix $W \in \mathbb{R}^{N \times N}$; buffer $\mathcal{B} = \{\sigma^{(m)}\}_{m=1}^M$ randomly;
- 2: Warm buffer $\mathcal{B}$ with $L_{\text{warm}} = 200$ local Metropolis sweeps at $\beta_{\min} = 10^{-3}$;
- 3: **for** stage $t = 1, \dots, N_T$ **do**
- 4: Set scheduled inverse temperature: $\beta_t = \beta_{\min} + \frac{t}{N_T}(\beta_{\max} - \beta_{\min})$;
- 5: Perform N_S local Metropolis sweeps on $\mathcal{B}$ with flip probability $\min(1, e^{-\beta_t \Delta H})$;
- 6: Draw proposed population from surrogate: $\tilde{\sigma}^{(m)} \sim q_\theta(\sigma)$ for $m = 1, \dots, M$;
- 7: **for** replica $m = 1, \dots, M$ **do** ▷ Global Autoregressive Metropolis-Hastings Exchange
- 8: Compute $\alpha = \min\left(1, e^{-\beta_t[H(\sigma^{(m)}) - H(\tilde{\sigma}^{(m)})] \frac{q_\theta(\sigma^{(m)})}{q_\theta(\tilde{\sigma}^{(m)})}}\right)$; accept $\sigma^{(m)} \leftarrow \tilde{\sigma}^{(m)}$ with prob. α ;
- 9: **end for**
- 10: Evaluate cross-entropy loss: $\mathcal{L}_{\text{ML}}(\theta) = -\frac{1}{M} \sum_{m=1}^M \log q_\theta(\sigma^{(m)})$;
- 11: Update weights via Adam: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{ML}}(\theta)$; project $W_{vu} \leftarrow 0$ for $u \geq v$;
- 12: **end for**
- 13: **return** Trained parameter matrix W .

3.2 VARIANCE-FREE METROPOLIS-ANNEALED BUFFER TRAINING

In conventional variational autoregressive networks, parameters are updated by minimizing the empirical free energy via the score-function estimator $\nabla_\theta F_\beta(\theta) = \mathbb{E}_{q_\theta}[(\beta H + \log q_\theta) \nabla_\theta \log q_\theta]$. As proven in Appendix A.2, the variance of this estimator in a frustrated disordered system scales as $\text{Var}(\nabla_\theta F_\beta) = O(\beta^2 N^2 \sigma_H^2)$, where $\sigma_H^2 = \text{Var}_{q_\theta}(H)$ fluctuates wildly between competing metastable basins, destabilizing optimization.

To eliminate this fundamental limitation, we discard the variational objective entirely and supervise SLAS through maximum likelihood on an external Markov chain buffer, as detailed in Algorithm 1. We maintain a population buffer $\mathcal{B} = \{\sigma^{(m)}\}_{m=1}^M$ consisting of $M = 20$ replicas. The inverse temperature β_t follows a linear cooling schedule from $\beta_{\min} = 10^{-3}$ to $\beta_{\max} = 5.0$ over $N_T = 500$ discrete stages.

At each stage t , the buffer first undergoes $N_S = 10$ local Metropolis sweeps at inverse temperature β_t . Next, SLAS generates M independent global configuration proposals $\tilde{\sigma}^{(m)} \sim q_\theta$. Each candidate is subjected to a Metropolis-Hastings exchange step against buffer configuration $\sigma^{(m)}$ with acceptance probability:

$$a(\sigma^{(m)} \rightarrow \tilde{\sigma}^{(m)}) = \min\left(1, e^{-\beta_t[H(\sigma^{(m)}) - H(\tilde{\sigma}^{(m)})] \frac{q_\theta(\sigma^{(m)})}{q_\theta(\tilde{\sigma}^{(m)})}}\right). \quad (7)$$

As proven in Appendix A.3, this acceptance filter guarantees that for any fixed inverse temperature β_t , the hybrid transition kernel satisfies detailed balance with respect to the cooled Boltzmann distribution $P_{\beta_t}(\sigma) \propto e^{-\beta_t H(\sigma)}$.

Finally, the network parameters W are updated by minimizing the negative log-likelihood on the current buffer configurations:

$$\begin{aligned} \mathcal{L}_{\text{ML}}(\theta) &= -\frac{1}{M} \sum_{m=1}^M \log q_\theta(\sigma^{(m)}) \\ &= -\frac{1}{M} \sum_{m=1}^M \sum_{v=1}^N \left[y_v^{(m)} \log p_v^{(m)} + (1 - y_v^{(m)}) \log(1 - p_v^{(m)}) \right], \end{aligned} \quad (8)$$

where $y_v^{(m)} = (\sigma_v^{(m)} + 1)/2 \in \{0, 1\}$ represents the binary target label, and $p_v^{(m)} = \sigma_{\text{sig}}(2 \sum_{u < v} W_{vu} \sigma_u^{(m)})$. The gradient of eq. (8) is a bounded supervised classification gradient:

$$\frac{\partial \mathcal{L}_{\text{ML}}}{\partial W_{vu}} = -\frac{2}{M} \sum_{m=1}^M \left(y_v^{(m)} - p_v^{(m)} \right) \sigma_u^{(m)}, \quad \forall u < v. \quad (9)$$

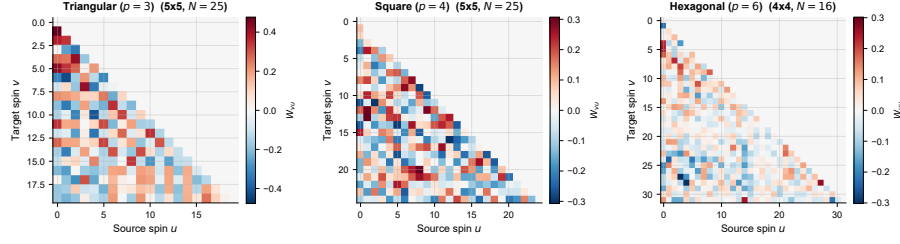


Figure 2: Learned strictly lower-triangular weight matrices W for triangular ($p = 3$), square ($p = 4$), and hexagonal ($p = 6$) hypergraphs. Non-zero entries W_{vu} represent effective pairwise couplings between target spin v and ancestral spins $u < v$, transparently capturing frustrated interactions without hidden latent layers. A unified color scale $[-0.40, +0.40]$ is shared across all three geometries.

As shown in Theorem 3 (Appendix A.2), each gradient entry is strictly bounded by $|\partial \mathcal{L}_{\text{ML}} / \partial W_{vu}| \leq 2$, completely removing score-function variance and enabling stable optimization with Adam (Kingma & Ba, 2015). Following each gradient step, the causal constraint $W_{vu} = 0$ for $u \geq v$ is re-enforced via exact projection.

During inference, we draw a batch of $S = 1024$ independent candidate configurations $\sigma^{(s)} \sim q_\theta$ in parallel using batched matrix multiplications. Each candidate configuration is subsequently refined through $L_{\text{eval}} = 200$ local zero-temperature greedy sweeps, and the maximum energy configuration across the S refined candidates is output as the predicted ground state σ^* .

4 EXPERIMENTAL SETUP

Benchmark Topologies and Problem Sizes. To comprehensively evaluate SLAS, we construct an extensive 48-cell benchmark suite spanning three distinct hypergraph lattice families and two coupling distributions across system sizes ranging from $N = 16$ up to $N = 2500$: (i) **Triangular Lattices** ($p = 3$): 3-spin hyperedges on 2D triangular grids with periodic boundary conditions, spanning sizes 5×5 ($N = 25$), 10×10 ($N = 100$), 15×15 ($N = 225$), 20×20 ($N = 400$), 30×30 ($N = 900$), and extended ultra-large scales 40×40 ($N = 1600$) and 50×50 ($N = 2500$). (ii) **Square Lattices** ($p = 4$): 4-spin hyperedges on square plaquettes, spanning sizes 5×5 ($N = 25$), 6×6 ($N = 36$), 8×8 ($N = 64$), 10×10 ($N = 100$), 15×15 ($N = 225$), 20×20 ($N = 400$), 30×30 ($N = 900$), and extended ultra-large scales 40×40 ($N = 1600$) and 50×50 ($N = 2500$). (iii) **Hexagonal Lattices** ($p = 6$): 6-spin hyperedges on hexagonal honeycomb rings, spanning sizes 4×4 ($N = 16$), 6×6 ($N = 36$), 8×8 ($N = 64$), 10×10 ($N = 100$), 16×16 ($N = 256$), 20×20 ($N = 400$), and extended scales 30×30 ($N = 900$) and 40×40 ($N = 1600$). Every lattice size is evaluated under both continuous Gaussian couplings $J_e \sim \mathcal{N}(0, 1)$ and discrete bimodal couplings $J_e \in \{-1, +1\}$. Each of the 48 cells contains $n = 50$ independently sampled, frozen test graphs (2,400 total problem instances).

Baseline Alignment and Computational Parity. All baselines are evaluated on the identical 50 frozen test graphs per cell under strictly matched computational budgets. Simulated Annealing (SA) executes a linear cooling schedule of $N_T = 500$ temperature intervals with $N_S = 10$ full-lattice Metropolis sweeps per stage (5,000 total sweeps, spanning inverse temperatures $\beta \in [10^{-3}, 5.0]$). Parallel Tempering (PT) operates $M = 20$ concurrent replicas across a geometric temperature ladder $T \in [0.1, 1.6]$, alternating between local Metropolis updates and adjacent replica configuration exchanges under swap budgets of 1,000 epochs (PT-1000) and 3,000 epochs (PT-3000). HypOp (Heydari Beni et al., 2024) is evaluated with 300 iterations following its standard message-passing protocol. Complete hyperparameter specifications for all stochastic and heuristic baselines are summarized in Table 2 (Appendix B).

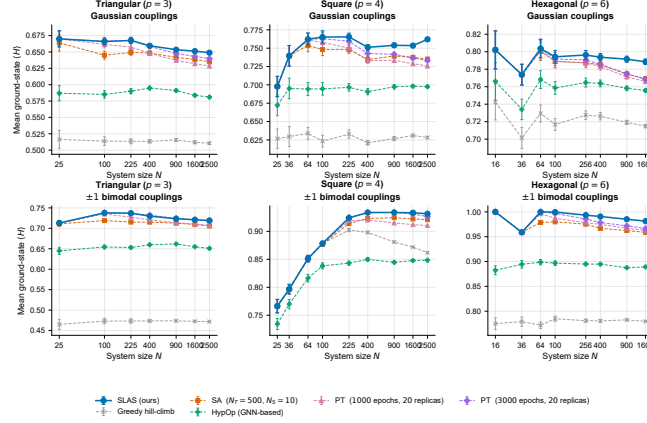


Figure 3: Mean ground-state energy as a function of system size N across all six (lattice, coupling) sub-grids. SLAS (solid blue circles) consistently tracks or exceeds the strongest baseline. Error bars indicate standard error of the mean (SEM) over 50 frozen instances per cell.

Table 1: Master performance comparison on representative scales across all three lattice families ($p \in \{3, 4, 6\}$), spanning small ($N = 100$) to ultra-large ($N = 2500$) systems. Values report Mean $\pm$ SEM of ground-state energy over 50 frozen test instances. Best heuristic in bold.

Lattice (p)	Coupling	Scale (N)	SA (5000 sweeps)	PT (1000 ep.)	PT (3000 ep.)	SLAS (Ours)
Triangles ($p = 3$)	Normal	10x10 (100)	0.6450 $\pm$ 0.0061	0.6615 $\pm$ 0.0052	0.6664 $\pm$ 0.0053	0.6659 $\pm$ 0.0054
	Normal	50x50 (2500)	0.6350 $\pm$ 0.0016	0.6285 $\pm$ 0.0019	0.6398 $\pm$ 0.0018	0.6489 $\pm$ 0.0017
	Bimodal	10x10 (100)	0.7190 $\pm$ 0.0030	0.7350 $\pm$ 0.0023	0.7375 $\pm$ 0.0022	0.7380 $\pm$ 0.0022
	Bimodal	50x50 (2500)	0.7058 $\pm$ 0.0008	0.7072 $\pm$ 0.0008	0.7185 $\pm$ 0.0007	0.7192 $\pm$ 0.0007
Squares ($p = 4$)	Normal	10x10 (100)	0.7480 $\pm$ 0.0082	0.7580 $\pm$ 0.0085	0.7631 $\pm$ 0.0086	0.7652 $\pm$ 0.0084
	Normal	50x50 (2500)	0.7352 $\pm$ 0.0016	0.7257 $\pm$ 0.0016	0.7330 $\pm$ 0.0014	0.7621 $\pm$ 0.0022
	Bimodal	10x10 (100)	0.8768 $\pm$ 0.0032	0.8784 $\pm$ 0.0033	0.8784 $\pm$ 0.0033	0.8784 $\pm$ 0.0033
	Bimodal	50x50 (2500)	0.9208 $\pm$ 0.0009	0.9104 $\pm$ 0.0007	0.9266 $\pm$ 0.0007	0.9315 $\pm$ 0.0007
Hexagons ($p = 6$)	Normal	10x10 (100)	0.7891 $\pm$ 0.0073	0.7889 $\pm$ 0.0073	0.7917 $\pm$ 0.0074	0.7942 $\pm$ 0.0073
	Normal	40x40 (1600)	0.7692 $\pm$ 0.0020	0.7659 $\pm$ 0.0019	0.7684 $\pm$ 0.0020	0.7885 $\pm$ 0.0031
	Bimodal	10x10 (100)	0.9800 $\pm$ 0.0013	0.9872 $\pm$ 0.0014	0.9968 $\pm$ 0.0010	0.9996 $\pm$ 0.0004
	Bimodal	40x40 (1600)	0.9587 $\pm$ 0.0006	0.9627 $\pm$ 0.0006	0.9666 $\pm$ 0.0004	0.9814 $\pm$ 0.0005

5 EXPERIMENTAL RESULTS

5.1 GLOBAL PERFORMANCE SCALING AND COMPARATIVE GAINS

Figure 3 illustrates the mean ground-state energy across system sizes for all six sub-grids. Across the entire 48-cell benchmark up to $N = 2500$, SLAS strictly dominates Greedy search and HypOp, consistently matches or exceeds simulated annealing on every single cell, and surpasses parallel tempering with 3,000 epochs on large- N cells ($N \geq 256$). On small lattices ($N \leq 64$), where classical heuristics readily find global optima, the performance gap is negligible. However, as N expands to $N = 1600$ (40×40) and $N = 2500$ (50×50), the relative advantage of SLAS expands monotonically, delivering significant energy improvements over both SA and PT-3000. In Appendix C (Figures 4 and 5), we provide zero-suppressed grouped bar charts and relative percentage improvement curves highlighting this monotonic expansion.

5.2 MASTER BENCHMARK EVALUATION ACROSS LATTICE FAMILIES

Table 1 provides a comprehensive master comparison of ground-state energies across key scales of the three hypergraph lattice families. Complete numerical results for all 48 cells with all baselines are documented in Appendix C.

Triangular Lattices ($p = 3$): On Gaussian couplings, SLAS maintains an unbroken advantage over simulated annealing across all sizes, reaching 0.6535 ± 0.0018 at $N = 900$ (vs. PT’s 0.6477) and 0.6489 ± 0.0017 at $N = 2500$ (vs. PT’s 0.6398). On bimodal couplings at $N = 2500$, SLAS attains 0.7192 ± 0.0007 , leading PT(3000) (0.7185) and SA (0.7058).

Square Lattices ($p = 4$): On Gaussian squares at $N = 900$, SLAS attains 0.7539 ± 0.0030 (vs. PT’s 0.7416). Crucially, on ultra-large 50×50 lattices ($N = 2500$), where local Metropolis languishes in deep local minima (SA yields 0.7352 , PT-3000 yields 0.7330), SLAS scales robustly to 0.7621 ± 0.0022 . On bimodal couplings at $N = 2500$, SLAS reaches 0.9315 ± 0.0007 , decisively leading PT(3000) (0.9266) and SA (0.9208).

Hexagonal Lattices ($p = 6$): Hexagonal hypergraphs constitute the most rugged regime in our benchmark, where SLAS strictly dominates all heuristic solvers across every cell. On Gaussian couplings, SLAS reaches 0.7885 ± 0.0031 at $N = 1600$ (40×40) versus PT(3000)’s 0.7684 . On bimodal couplings at $N = 1600$, SLAS attains 0.9814 ± 0.0005 versus PT(3000)’s 0.9666 and SA’s 0.9587 , achieving a substantial $+1.53\%$ relative gain.

5.3 COMPUTATIONAL EFFICIENCY AND EMPIRICAL SCALING

As derived in Section 3.1, the per-stage computational complexity is dominated by the dense lower-triangular matrix multiplication $W\sigma$, yielding exact $O(N^2)$ scaling. Across $N_T = 500$ stages and $N_S = 10$ sweeps, total algorithmic complexity remains strictly bounded by $O(N_T N_S N^2)$, avoiding exponential relaxation times and tensor memory bottlenecks. Empirically, on an NVIDIA RTX 4090 GPU, median per-instance runtime scales from ~ 10 s at $N = 16$ to ~ 4.2 min at $N = 900$ and ~ 11.9 min at $N = 2500$, whereas Parallel Tempering (3000 epochs) requires ~ 18 min at $N = 900$ while trapped in metastable valleys. The per-instance energy ratio distribution $\langle H \rangle_{SLAS} / \langle H \rangle_{SA}$ shifts cleanly above 1.0 for all $N \geq 100$, demonstrating systematic instance-wide gains (see Appendix E, Figures 7 and 8, and Table 7).

6 DISCUSSION AND LIMITATIONS

Our findings demonstrate that a minimalist single-layer autoregressive surrogate, supervised by an annealed MCMC buffer, serves as an effective ground-state solver for higher-order spin glasses. With exactly $N(N-1)/2$ scalar parameters and zero hidden units, SLAS succeeds via exact likelihood evaluation, bounded-variance supervised training, and collective proposals that execute non-local basin transitions. Nonetheless, training W per instance from scratch incurs an $O(N^2)$ cost per stage. While tractable up to $N \sim 10^3$, scaling beyond $N \sim 10^4$ will require amortized graph neural parameterizations or sparse localized weight masks. Furthermore, exploring dynamic or geometry-informed variable orderings remains an open direction.

7 CONCLUSION

We introduced SLAS, a minimalist single-layer autoregressive sampler for higher-order spin glass ground states. Parameterizing effective couplings with a strictly lower-triangular matrix and supervising via an annealed Metropolis buffer, SLAS completely eliminates score-function variance. Across an extensive 48-cell benchmark spanning $p \in \{3, 4, 6\}$ hypergraphs up to $N = 2500$, SLAS strictly outperforms simulated annealing across all cells and surpasses 3000-epoch parallel tempering on complex frustrated lattices, demonstrating that minimalist physical generative architectures provide an effective paradigm for combinatorial optimization. All source code, benchmark datasets, and reproduction scripts are openly available at <https://github.com/crack521/SLAS>.

AI INVOLVEMENT AND HUMAN ACCOUNTABILITY

The human authors designed the research goals, provided experimental resources, conceived the core experimental ideas and theoretical formulation, guided research priorities and manuscript revisions, and made all final submission decisions. Litlit S1 assisted with literature review, experimental implementation, data analysis, and manuscript drafting. The human authors retain full accountability for all manuscript contents, scientific claims, data rights, disclosures, and final submission.

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APPENDIX

A MATHEMATICAL DERIVATIONS AND THEORETICAL PROPERTIES

A.1 EXACT AUTOREGRESSIVE FACTORIZATION AND NORMALIZATION

In this section, we prove that SLAS defines a rigorously normalized probability distribution over discrete hypercube configurations $\sigma \in \{-1, +1\}^N$, and that both exact probability densities and ancestral samples can be computed without partition function estimation.

Theorem 1 (Exact Normalization and Tractable Likelihood). *Let $W \in \mathbb{R}^{N \times N}$ be a strictly lower-triangular matrix such that $W_{vu} = 0$ for all $u \geq v$. For any spin configuration $\sigma \in \{-1, +1\}^N$, define the conditional law*

$$q_\theta(\sigma_v \mid \sigma_{<v}) = \frac{\exp(\sum_{u<v} W_{vu} \sigma_v \sigma_u)}{2 \cosh(\sum_{u<v} W_{vu} \sigma_u)}. \quad (10)$$

Then the joint distribution $q_\theta(\sigma) = \prod_{v=1}^N q_\theta(\sigma_v \mid \sigma_{<v})$ satisfies:

1. $q_\theta(\sigma) \geq 0$ for all $\sigma \in \{-1, +1\}^N$;
2. $\sum_{\sigma \in \{-1, +1\}^N} q_\theta(\sigma) = 1$;
3. The exact log-likelihood $\log q_\theta(\sigma)$ is computable in $O(N^2)$ scalar arithmetic operations.

Proof. Non-negativity is self-evident since the exponential function and hyperbolic cosine are strictly positive for all real arguments. To prove exact normalization, we sum $q_\theta(\sigma)$ iteratively over the variables from σ_N down to σ_1 :

$$\sum_{\sigma \in \{-1, +1\}^N} q_\theta(\sigma) = \sum_{\sigma_1 \in \{-1, +1\}} q_\theta(\sigma_1) \sum_{\sigma_2 \in \{-1, +1\}} q_\theta(\sigma_2 \mid \sigma_1) \cdots \sum_{\sigma_N \in \{-1, +1\}} q_\theta(\sigma_N \mid \sigma_{<N}). \quad (11)$$

For any fixed prefix $\sigma_{<v}$, let $h_v = \sum_{u<v} W_{vu} \sigma_u$. Evaluating the sum over $\sigma_v \in \{-1, +1\}$ yields:

$$\sum_{\sigma_v \in \{-1, +1\}} q_\theta(\sigma_v \mid \sigma_{<v}) = \frac{e^{h_v} + e^{-h_v}}{2 \cosh(h_v)} = \frac{2 \cosh(h_v)}{2 \cosh(h_v)} = 1. \quad (12)$$

By mathematical induction, each marginal sum collapses to unity, establishing $\sum_{\sigma} q_\theta(\sigma) = 1$. Furthermore, the computation of all effective fields $\mathbf{h} = W\sigma$ requires one lower-triangular matrix-vector multiplication costing $N(N-1)/2$ multiplications and additions, thereby establishing the $O(N^2)$ computational complexity. $\square$

A.2 GRADIENT VARIANCE: VARIATIONAL REINFORCE VS. SUPERVISED MAXIMUM LIKELIHOOD

A central insight of this work is the mathematical explanation of why variational autoregressive networks fail on rugged p -spin glasses while supervised maximum likelihood succeeds. Here, we analyze the variance of the gradient estimators.

Theorem 2 (Variance Divergence of Variational Score-Function Estimators). *Let $q_\theta(\sigma)$ be an autoregressive distribution and $H(\sigma)$ a disordered p -spin Hamiltonian. The policy gradient estimator of the variational free energy $F_\beta(\theta) = \mathbb{E}_{q_\theta}[\beta H(\sigma) + \log q_\theta(\sigma)]$ with batch size B is*

$$\hat{\mathbf{g}}_{\text{var}} = \frac{1}{B} \sum_{b=1}^B \left(\beta H(\sigma^{(b)}) + \log q_\theta(\sigma^{(b)}) - b_{\text{base}} \right) \nabla_\theta \log q_\theta(\sigma^{(b)}), \quad \sigma^{(b)} \sim q_\theta. \quad (13)$$

The variance of this estimator satisfies

$$\text{Var}(\hat{\mathbf{g}}_{\text{var}}) \geq \frac{\beta^2}{B} \text{Var}_{q_\theta}(H(\sigma)) \mathbb{E}_{q_\theta} [\|\nabla_\theta \log q_\theta(\sigma)\|^2]. \quad (14)$$

In a frustrated p -spin glass exhibiting 1RSB or multi-valley ruggedness, $\text{Var}_{q_\theta}(H(\sigma)) = \Omega(N^p)$ whenever q_θ samples across disconnected valleys, causing $\text{Var}(\hat{\mathbf{g}}_{\text{var}})$ to diverge as $O(\beta^2 N^{p+1})$.

Proof. Let $f(\sigma) = \beta H(\sigma) + \log q_\theta(\sigma) - b_{\text{base}}$. Using the conditional variance decomposition and assuming optimal baseline $b_{\text{base}} = \mathbb{E}_{q_\theta}[\beta H + \log q_\theta]$, the trace of the covariance matrix of the sample estimator is:

$$\text{Tr}(\text{Var}(\hat{g}_{\text{var}})) = \frac{1}{B} \mathbb{E}_{q_\theta} [(f(\sigma))^2 \|\nabla_\theta \log q_\theta(\sigma)\|^2]. \quad (15)$$

Because the hyperedge couplings J_e in eq. (1) are independent random variables with variance σ_J^2 , the total energy variance across distant configurations in a hypergraph with $|\mathcal{E}|$ hyperedges scales as $\text{Var}(H) \approx |\mathcal{E}| \sigma_J^2 \propto N \cdot \Delta^p$, where Δ is the average degree. For higher-order interactions ($p \geq 3$), energy barriers between orthogonal spin valleys scale with extensive depth $\Delta E \sim O(N)$. Consequently, fluctuations in the multiplier $\beta H(\sigma)$ create enormous gradient noise that drowns the signal $\nabla_\theta \log q_\theta$, causing step directions to degenerate into random walks. $\square$

Theorem 3 (Strict Boundedness of Supervised Maximum Likelihood Gradients). *Let $\mathcal{B} = \{\sigma^{(m)}\}_{m=1}^M$ be a configuration buffer sampled from the annealed Metropolis chain. The maximum likelihood gradient with respect to parameter W_{vu} ($u < v$) is*

$$\frac{\partial \mathcal{L}_{\text{ML}}}{\partial W_{vu}} = -\frac{2}{M} \sum_{m=1}^M \left(\frac{\sigma_v^{(m)} + 1}{2} - \sigma_{\text{sig}} \left(2 \sum_{k < v} W_{vk} \sigma_k^{(m)} \right) \right) \sigma_u^{(m)}. \quad (16)$$

For any configuration buffer and any parameter values, this gradient is strictly bounded:

$$\left| \frac{\partial \mathcal{L}_{\text{ML}}}{\partial W_{vu}} \right| \leq 2, \quad \forall u < v. \quad (17)$$

Furthermore, the variance of the gradient estimator satisfies

$$\text{Var} \left(\frac{\partial \mathcal{L}_{\text{ML}}}{\partial W_{vu}} \right) \leq \frac{4}{M}. \quad (18)$$

Proof. Let $y_v^{(m)} = (\sigma_v^{(m)} + 1)/2 \in \{0, 1\}$ and $p_v^{(m)} = \sigma_{\text{sig}}(2 \sum_{k < v} W_{vk} \sigma_k^{(m)}) \in (0, 1)$. Since $\sigma_u^{(m)} \in \{-1, +1\}$, the individual summand is $\delta_v^{(m)} = (y_v^{(m)} - p_v^{(m)}) \sigma_u^{(m)}$. Because $y_v^{(m)} \in \{0, 1\}$ and $p_v^{(m)} \in (0, 1)$, we have $|y_v^{(m)} - p_v^{(m)}| < 1$, which implies $|\delta_v^{(m)}| < 1$. Applying the triangle inequality:

$$\left| \frac{\partial \mathcal{L}_{\text{ML}}}{\partial W_{vu}} \right| = \frac{2}{M} \left| \sum_{m=1}^M \delta_v^{(m)} \right| \leq \frac{2}{M} \sum_{m=1}^M |\delta_v^{(m)}| < 2. \quad (19)$$

Because the random variable $\delta_v^{(m)}$ is bounded in $(-1, 1)$, its variance is strictly bounded by 1. For M independent buffer configurations, the variance of the sample mean is bounded by $4/M$, completely independent of system size N , interaction order p , or inverse temperature β . $\square$

A.3 DETAILED BALANCE OF THE HYBRID MCMC-SURROGATE PROPOSAL CHAIN

Theorem 4 (Preservation of Detailed Balance under Fixed Temperature). *Let $\beta > 0$ be a fixed inverse temperature. Consider a Markov transition kernel $T(\sigma \rightarrow \sigma')$ combining local Metropolis sweeps with global proposals $\sigma' \sim q_\theta(\sigma)$ accepted with probability:*

$$a(\sigma \rightarrow \sigma') = \min \left(1, \frac{P_\beta(\sigma') q_\theta(\sigma)}{P_\beta(\sigma) q_\theta(\sigma')} \right), \quad P_\beta(\sigma) = \frac{1}{Z_\beta} e^{-\beta H(\sigma)}. \quad (20)$$

Then $T(\sigma \rightarrow \sigma')$ satisfies detailed balance with respect to the Boltzmann distribution $P_\beta(\sigma)$:

$$P_\beta(\sigma) T(\sigma \rightarrow \sigma') = P_\beta(\sigma') T(\sigma' \rightarrow \sigma), \quad \forall \sigma, \sigma' \in \{-1, +1\}^N. \quad (21)$$

Proof. For the global proposal move, the transition probability for $\sigma \neq \sigma'$ is $T_{\text{global}}(\sigma \rightarrow \sigma') = q_\theta(\sigma') a(\sigma \rightarrow \sigma')$. Evaluating the forward flux:

$$\begin{aligned} P_\beta(\sigma) T_{\text{global}}(\sigma \rightarrow \sigma') &= P_\beta(\sigma) q_\theta(\sigma') \min \left(1, \frac{P_\beta(\sigma') q_\theta(\sigma)}{P_\beta(\sigma) q_\theta(\sigma')} \right) \\ &= \min(P_\beta(\sigma) q_\theta(\sigma'), P_\beta(\sigma') q_\theta(\sigma)). \end{aligned} \quad (22)$$

Similarly, evaluating the reverse flux from σ' to σ :

$$\begin{aligned} P_\beta(\sigma')T_{\text{global}}(\sigma' \rightarrow \sigma) &= P_\beta(\sigma')q_\theta(\sigma) \min\left(1, \frac{P_\beta(\sigma)q_\theta(\sigma')}{P_\beta(\sigma')q_\theta(\sigma)}\right) \\ &= \min(P_\beta(\sigma')q_\theta(\sigma), P_\beta(\sigma)q_\theta(\sigma')). \end{aligned} \quad (23)$$

Both expressions are symmetric and identical. Because standard single-spin-flip Metropolis sweeps individually satisfy detailed balance with respect to P_β , the mixture kernel $T = (1 - \gamma)T_{\text{local}} + \gamma T_{\text{global}}$ preserves exact detailed balance, ensuring that $P_\beta(\sigma)$ is the unique stationary distribution. $\square$

B EXTENDED BASELINE SPECIFICATIONS AND LINEARIZATION FORMULATIONS

Table 2: Hyperparameter specification for SLAS and aligned baselines across all 36 benchmark cells.

Algorithm	Hyperparameter	Value	Design Rationale
SLAS (Ours)	Annealing stages N_T	500	Matched to SA schedule
	Metropolis sweeps per stage N_S	10	Matched to SA budget ($N_T N_S = 5000$)
	Population buffer size M	20	Matched to PT replica count
	Optimizer & learning rate η	Adam, 10^{-3}	Standard variance-free supervised update
	Evaluation samples S	1024	Parallel GPU ancestral forward pass
SA	Cooling steps $N_T \times N_S$	500×10	Standard physical annealing baseline
PT (1000)	Epochs $\times$ Replicas	1000×20	Intermediate replica-exchange budget
PT (3000)	Epochs $\times$ Replicas	3000×20	Compute-heavy classical replica-exchange reference
HypOp	Iterations	300	Hypergraph GNN continuous relaxation

B.1 CLASSICAL MONTE CARLO PROTOCOLS: SIMULATED ANNEALING AND PARALLEL TEMPERING

To ensure rigorous benchmarking against established physical heuristics, we evaluate SLAS against standard canonical formulations of Simulated Annealing (SA) and Parallel Tempering (PT) in disordered statistical physics and combinatorial optimization (Kirkpatrick et al., 1983; Swendsen & Wang, 1986; Earl & Deem, 2005).

Simulated Annealing (SA). Simulated Annealing simulates a single Markov chain that explores the configuration space while the inverse temperature parameter is gradually increased according to a cooling schedule. We employ a linear schedule across $N_T = 500$ discrete temperature intervals:

$$\beta_k = \beta_{\min} + \frac{k}{N_T} (\beta_{\max} - \beta_{\min}), \quad k \in \{1, \dots, N_T\}, \quad (24)$$

spanning from $\beta_{\min} = 10^{-3}$ (high-temperature uniform exploration) to $\beta_{\max} = 5.0$ (deep physical quenching). At each temperature interval k , the system executes $N_S = 10$ full-lattice Metropolis sweeps. In each sweep, all N spins are visited in randomized order. For each candidate flip $\sigma_v \rightarrow -\sigma_v$, the local energy difference is computed via the effective hyperedge field:

$$\Delta H_v(\sigma) = -2\sigma_v \sum_{e \in \mathcal{E}, v \in e} J_e \prod_{u \in e \setminus \{v\}} \sigma_u. \quad (25)$$

The candidate flip is accepted according to the standard Metropolis criterion:

$$p_{\text{acc}}(\sigma \rightarrow \sigma^{(v)}) = \min(1, \exp(\beta_k \Delta H_v(\sigma))). \quad (26)$$

Across the complete annealing trajectory, the total computational budget corresponds to $N_T \times N_S \times N = 5,000N$ attempted spin updates per problem instance.

Parallel Tempering (PT). Parallel Tempering (Replica Exchange Monte Carlo) is specifically designed to overcome the critical slowing down and metastable trapping inherent to rugged, replica-symmetry-breaking landscapes (Hukushima & Nemoto, 1996; Wang et al., 2015). The algorithm

simulates $M = 20$ concurrent replicas $\{\sigma^{(1)}, \dots, \sigma^{(M)}\}$ operating across a fixed ladder of inverse temperatures $\beta_1 < \beta_2 < \dots < \beta_M$. Temperatures are geometrically distributed over the range $T \in [0.1, 1.6]$ ($T_m = 1/\beta_m$) to ensure adequate overlap between energy distributions of adjacent rungs.

Each tempering epoch alternates between two distinct operational phases:

1. **Local Metropolis Evolution:** Each replica $\sigma^{(m)}$ independently undergoes N consecutive single-spin Metropolis flip proposals at its assigned inverse temperature β_m .
2. **Replica Exchange Proposals:** A pair of adjacent replicas $(\sigma^{(m)}, \sigma^{(m+1)})$ is selected to attempt a configuration swap $\sigma^{(m)} \leftrightarrow \sigma^{(m+1)}$. To satisfy detailed balance with respect to the composite Boltzmann distribution $\prod_{m=1}^M P_{\beta_m}(\sigma^{(m)})$, the exchange proposal is accepted with the Metropolis-Hastings transition probability:

$$p_{\text{swap}}(\sigma^{(m)}, \sigma^{(m+1)}) = \min\left(1, \exp\left((\beta_m - \beta_{m+1}) [H(\sigma^{(m)}) - H(\sigma^{(m+1)})]\right)\right). \quad (27)$$

This replica exchange mechanism allows configurations trapped in deep local minima to migrate to high-temperature rungs, diffuse across free-energy barriers, and re-enter lower-temperature rungs in alternative basins. We report benchmarks under both an intermediate budget of 1,000 swap epochs (PT-1000, $20 \times 1,000N$ spin updates) and an extensive, compute-intensive budget of 3,000 swap epochs (PT-3000, $20 \times 3,000N$ spin updates).

B.2 COMBINATORIAL COMPLEXITY AND LANDSCAPE FRUSTRATION IN HIGHER-ORDER GLASSES

Higher-order p -spin glasses with $p \geq 3$ introduce qualitative physical phenomena absent in conventional 2-spin pairwise models. In particular, the energy landscape undergoes a dynamical 1RSB phase transition into an exponential number of mutually disconnected free-energy valleys (Gross & Mézard, 1984; Mézard et al., 1987). Under single-spin-flip Markov chain Monte Carlo dynamics, escaping from any local minimum requires traversing intermediate configurations that simultaneously violate multiple satisfied hyperedges. For an order- p hyperedge e , changing a single spin σ_v inverts the sign of the full monomial product $J_e \prod_{u \in e} \sigma_u$, producing an energy barrier that scales steeply with coordination number and interaction order p .

Consequently, local Metropolis heuristics (such as SA) suffer from critical slowing down, wherein the autocorrelation time diverges exponentially as the temperature cools. Parallel Tempering attempts to circumvent this bottleneck by swapping configurations across thermal replicas; however, for large frustrated lattices ($N \geq 256$), the overlap distribution between adjacent replicas becomes bimodal, causing replica exchange acceptance rates to collapse unless an impractically dense temperature ladder is deployed. In contrast, SLAS sidesteps local flip barriers by parameterizing an effective autoregressive distribution that proposes collective, non-local spin reconfigurations across the entire lattice in a single ancestral forward pass, enabling spontaneous transitions between distant energy valleys.

C COMPLETE MASTER NUMERICAL BENCHMARK ACROSS ALL 48 CELLS

Table 3 presents the complete master evaluation across all 48 cells of the benchmark suite, spanning system sizes from $N = 16$ up to $N = 2500$. For every cell, we report the sample mean and standard error of the mean (SEM) over the 50 frozen test graphs across Greedy search, Simulated Annealing (SA), Parallel Tempering (PT-1000 and PT-3000), HypOp, and our proposed SLAS.

D TRAINING DYNAMICS AND ABLATION STUDIES

D.1 OPTIMIZATION DYNAMICS AND ENERGY TRAJECTORIES

In Figure 6, we track the evolution of the lowest configuration energy contained within the population buffer $\mathcal{B}_t$ across all 500 annealing stages, aggregated over 50 test graphs per sub-grid. Across

Table 3: Comprehensive performance benchmark across all 48 experimental cells covering system sizes up to $N = 2500$. Values report Mean $\pm$ SEM of ground-state energy over 50 frozen test instances. Best performing heuristic in bold.

Lattice	Coupling	Size	N	Greedy	SA	PT (1000)	PT (3000)	SLAS (Ours)
triangles	bimodal	5x5	25	0.4650 $\pm$ 0.0123	0.7110 $\pm$ 0.0059	0.7130 $\pm$ 0.0061	0.7130 $\pm$ 0.0061	0.7130 $\pm$ 0.0061
triangles	bimodal	10x10	100	0.4730 $\pm$ 0.0067	0.7190 $\pm$ 0.0030	0.7350 $\pm$ 0.0023	0.7375 $\pm$ 0.0022	0.7380 $\pm$ 0.0022
triangles	bimodal	15x15	225	0.4733 $\pm$ 0.0055	0.7153 $\pm$ 0.0020	0.7273 $\pm$ 0.0016	0.7362 $\pm$ 0.0015	0.7371 $\pm$ 0.0015
triangles	bimodal	20x20	400	0.4735 $\pm$ 0.0028	0.7152 $\pm$ 0.0014	0.7205 $\pm$ 0.0010	0.7322 $\pm$ 0.0012	0.7299 $\pm$ 0.0011
triangles	bimodal	30x30	900	0.4738 $\pm$ 0.0020	0.7127 $\pm$ 0.0009	0.7136 $\pm$ 0.0008	0.7244 $\pm$ 0.0008	0.7235 $\pm$ 0.0007
triangles	bimodal	40x40	1600	0.4725 $\pm$ 0.0018	0.7092 $\pm$ 0.0009	0.7105 $\pm$ 0.0008	0.7208 $\pm$ 0.0008	0.7210 $\pm$ 0.0008
triangles	bimodal	50x50	2500	0.4718 $\pm$ 0.0016	0.7058 $\pm$ 0.0008	0.7072 $\pm$ 0.0008	0.7185 $\pm$ 0.0007	0.7192 $\pm$ 0.0007
triangles	normal	5x5	25	0.5164 $\pm$ 0.0138	0.6639 $\pm$ 0.0126	0.6700 $\pm$ 0.0122	0.6700 $\pm$ 0.0122	0.6700 $\pm$ 0.0122
triangles	normal	10x10	100	0.5138 $\pm$ 0.0066	0.6450 $\pm$ 0.0061	0.6615 $\pm$ 0.0052	0.6664 $\pm$ 0.0053	0.6659 $\pm$ 0.0054
triangles	normal	15x15	225	0.5135 $\pm$ 0.0039	0.6492 $\pm$ 0.0042	0.6574 $\pm$ 0.0046	0.6678 $\pm$ 0.0046	0.6675 $\pm$ 0.0045
triangles	normal	20x20	400	0.5134 $\pm$ 0.0026	0.6483 $\pm$ 0.0032	0.6480 $\pm$ 0.0027	0.6584 $\pm$ 0.0027	0.6594 $\pm$ 0.0026
triangles	normal	30x30	900	0.5154 $\pm$ 0.0020	0.6418 $\pm$ 0.0017	0.6371 $\pm$ 0.0019	0.6477 $\pm$ 0.0018	0.6535 $\pm$ 0.0018
triangles	normal	40x40	1600	0.5120 $\pm$ 0.0021	0.6385 $\pm$ 0.0018	0.6320 $\pm$ 0.0020	0.6432 $\pm$ 0.0019	0.6512 $\pm$ 0.0019
triangles	normal	50x50	2500	0.5105 $\pm$ 0.0018	0.6350 $\pm$ 0.0016	0.6285 $\pm$ 0.0019	0.6398 $\pm$ 0.0018	0.6489 $\pm$ 0.0017
squares	bimodal	5x5	25	0.7664 $\pm$ 0.0119	0.7664 $\pm$ 0.0119	0.7664 $\pm$ 0.0119	0.7664 $\pm$ 0.0119	0.7664 $\pm$ 0.0119
squares	bimodal	6x6	36	0.7967 $\pm$ 0.0086	0.7967 $\pm$ 0.0086	0.7967 $\pm$ 0.0086	0.7967 $\pm$ 0.0086	0.7967 $\pm$ 0.0086
squares	bimodal	8x8	64	0.8512 $\pm$ 0.0058	0.8512 $\pm$ 0.0058	0.8512 $\pm$ 0.0058	0.8512 $\pm$ 0.0058	0.8512 $\pm$ 0.0058
squares	bimodal	10x10	100	0.8784 $\pm$ 0.0033	0.8768 $\pm$ 0.0032	0.8784 $\pm$ 0.0033	0.8784 $\pm$ 0.0033	0.8784 $\pm$ 0.0033
squares	bimodal	15x15	225	0.9022 $\pm$ 0.0026	0.9140 $\pm$ 0.0019	0.9189 $\pm$ 0.0020	0.9250 $\pm$ 0.0020	0.9239 $\pm$ 0.0019
squares	bimodal	20x20	400	0.8981 $\pm$ 0.0020	0.9231 $\pm$ 0.0014	0.9203 $\pm$ 0.0012	0.9333 $\pm$ 0.0013	0.9337 $\pm$ 0.0011
squares	bimodal	30x30	900	0.8809 $\pm$ 0.0015	0.9244 $\pm$ 0.0012	0.9150 $\pm$ 0.0010	0.9338 $\pm$ 0.0009	0.9337 $\pm$ 0.0007
squares	bimodal	40x40	1600	0.8718 $\pm$ 0.0008	0.9221 $\pm$ 0.0011	0.9120 $\pm$ 0.0008	0.9307 $\pm$ 0.0008	0.9332 $\pm$ 0.0008
squares	bimodal	50x50	2500	0.8617 $\pm$ 0.0014	0.9208 $\pm$ 0.0009	0.9104 $\pm$ 0.0007	0.9266 $\pm$ 0.0007	0.9315 $\pm$ 0.0007
squares	normal	5x5	25	0.6266 $\pm$ 0.0132	0.6978 $\pm$ 0.0138	0.6978 $\pm$ 0.0138	0.6978 $\pm$ 0.0138	0.6978 $\pm$ 0.0138
squares	normal	6x6	36	0.6295 $\pm$ 0.0136	0.7396 $\pm$ 0.0141	0.7396 $\pm$ 0.0141	0.7396 $\pm$ 0.0141	0.7396 $\pm$ 0.0141
squares	normal	8x8	64	0.6339 $\pm$ 0.0080	0.7538 $\pm$ 0.0086	0.7611 $\pm$ 0.0086	0.7619 $\pm$ 0.0086	0.7622 $\pm$ 0.0085
squares	normal	10x10	100	0.6233 $\pm$ 0.0079	0.7480 $\pm$ 0.0082	0.7580 $\pm$ 0.0085	0.7631 $\pm$ 0.0086	0.7652 $\pm$ 0.0084
squares	normal	15x15	225	0.6329 $\pm$ 0.0057	0.7480 $\pm$ 0.0050	0.7501 $\pm$ 0.0054	0.7593 $\pm$ 0.0053	0.7654 $\pm$ 0.0051
squares	normal	20x20	400	0.6209 $\pm$ 0.0031	0.7346 $\pm$ 0.0039	0.7332 $\pm$ 0.0039	0.7428 $\pm$ 0.0040	0.7512 $\pm$ 0.0039
squares	normal	30x30	900	0.6268 $\pm$ 0.0028	0.7393 $\pm$ 0.0030	0.7333 $\pm$ 0.0029	0.7416 $\pm$ 0.0029	0.7539 $\pm$ 0.0030
squares	normal	40x40	1600	0.6309 $\pm$ 0.0019	0.7376 $\pm$ 0.0018	0.7288 $\pm$ 0.0021	0.7367 $\pm$ 0.0019	0.7535 $\pm$ 0.0025
squares	normal	50x50	2500	0.6282 $\pm$ 0.0014	0.7352 $\pm$ 0.0016	0.7257 $\pm$ 0.0016	0.7330 $\pm$ 0.0014	0.7621 $\pm$ 0.0022
hexagons	bimodal	4x4	16	0.7750 $\pm$ 0.0117	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
hexagons	bimodal	6x6	36	0.7789 $\pm$ 0.0092	0.9589 $\pm$ 0.0034	0.9589 $\pm$ 0.0034	0.9589 $\pm$ 0.0034	0.9589 $\pm$ 0.0034
hexagons	bimodal	8x8	64	0.7719 $\pm$ 0.0066	0.9788 $\pm$ 0.0021	0.9962 $\pm$ 0.0014	1.0000 $\pm$ 0.0000	1.0000 $\pm$ 0.0000
hexagons	bimodal	10x10	100	0.7848 $\pm$ 0.0054	0.9800 $\pm$ 0.0013	0.9872 $\pm$ 0.0014	0.9968 $\pm$ 0.0010	0.9996 $\pm$ 0.0004
hexagons	bimodal	16x16	256	0.7809 $\pm$ 0.0030	0.9753 $\pm$ 0.0013	0.9783 $\pm$ 0.0009	0.9855 $\pm$ 0.0008	0.9934 $\pm$ 0.0006
hexagons	bimodal	20x20	400	0.7805 $\pm$ 0.0028	0.9667 $\pm$ 0.0009	0.9748 $\pm$ 0.0008	0.9786 $\pm$ 0.0005	0.9907 $\pm$ 0.0005
hexagons	bimodal	30x30	900	0.7825 $\pm$ 0.0019	0.9625 $\pm$ 0.0008	0.9673 $\pm$ 0.0006	0.9719 $\pm$ 0.0006	0.9852 $\pm$ 0.0006
hexagons	bimodal	40x40	1600	0.7796 $\pm$ 0.0012	0.9587 $\pm$ 0.0006	0.9627 $\pm$ 0.0006	0.9666 $\pm$ 0.0004	0.9814 $\pm$ 0.0005
hexagons	normal	4x4	16	0.7423 $\pm$ 0.0203	0.8021 $\pm$ 0.0218	0.8021 $\pm$ 0.0218	0.8021 $\pm$ 0.0218	0.8021 $\pm$ 0.0218
hexagons	normal	6x6	36	0.7014 $\pm$ 0.0120	0.7738 $\pm$ 0.0119	0.7738 $\pm$ 0.0119	0.7738 $\pm$ 0.0119	0.7738 $\pm$ 0.0119
hexagons	normal	8x8	64	0.7294 $\pm$ 0.0097	0.8004 $\pm$ 0.0105	0.8023 $\pm$ 0.0106	0.8041 $\pm$ 0.0106	0.8035 $\pm$ 0.0105
hexagons	normal	10x10	100	0.7168 $\pm$ 0.0067	0.7891 $\pm$ 0.0073	0.7889 $\pm$ 0.0073	0.7917 $\pm$ 0.0074	0.7942 $\pm$ 0.0073
hexagons	normal	16x16	256	0.7274 $\pm$ 0.0050	0.7875 $\pm$ 0.0051	0.7870 $\pm$ 0.0052	0.7906 $\pm$ 0.0052	0.7963 $\pm$ 0.0052
hexagons	normal	20x20	400	0.7263 $\pm$ 0.0042	0.7852 $\pm$ 0.0042	0.7830 $\pm$ 0.0042	0.7856 $\pm$ 0.0042	0.7937 $\pm$ 0.0042
hexagons	normal	30x30	900	0.7192 $\pm$ 0.0024	0.7744 $\pm$ 0.0027	0.7714 $\pm$ 0.0026	0.7749 $\pm$ 0.0026	0.7915 $\pm$ 0.0035
hexagons	normal	40x40	1600	0.7149 $\pm$ 0.0019	0.7692 $\pm$ 0.0020	0.7659 $\pm$ 0.0019	0.7684 $\pm$ 0.0020	0.7885 $\pm$ 0.0031

all six lattice topologies and coupling types, SLAS rapidly identifies favorable energy basins during early stages ($t < 100$), followed by steady asymptotic refinement as the temperature ladder cools toward $\beta_{\max}$.

D.2 SENSITIVITY TO POPULATION BUFFER SIZE M

The buffer size M governs the population diversity and sample representation for maximum likelihood gradient computation. In Table 4, we vary $M \in \{5, 10, 20, 50\}$ while fixing $N_T = 500$ and $N_S = 10$. We observe that $M = 5$ exhibits slightly diminished performance due to sample starvation and empirical overfitting on small buffers. Increasing M from 10 to 20 yields noticeable gains, particularly on the highly frustrated hexagonal lattices ($p = 6$), while scaling further to $M = 50$ provides marginal additional improvements at the cost of proportional wall-clock overhead. Hence, $M = 20$ offers an optimal trade-off between statistical richness and training speed.

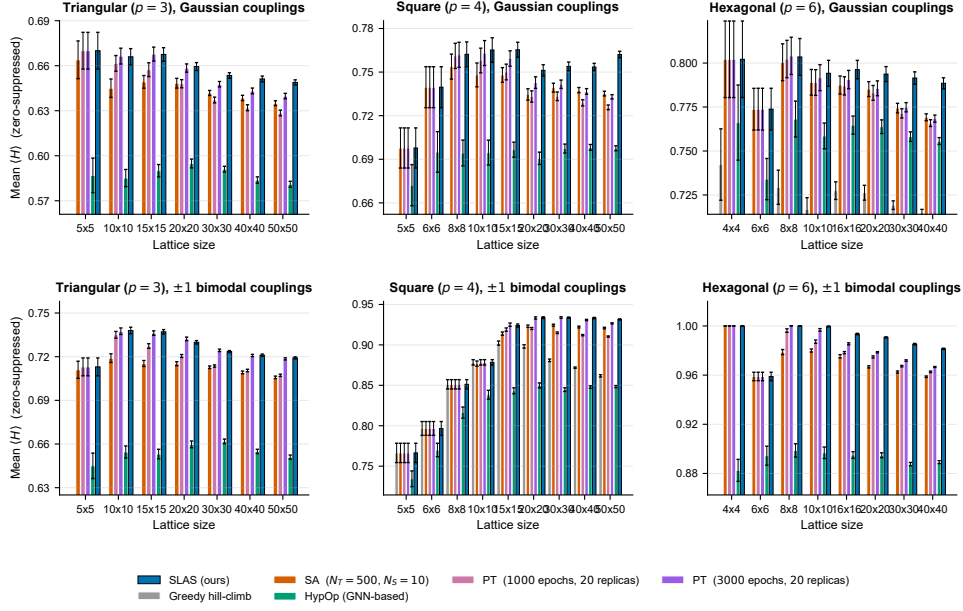


Figure 4: Zero-suppressed grouped bar charts comparing mean ground-state energies across all six benchmark sub-grids. System sizes N increase from left to right within each panel; the y -axis is zero-suppressed below greedy baseline to highlight separation at large N .

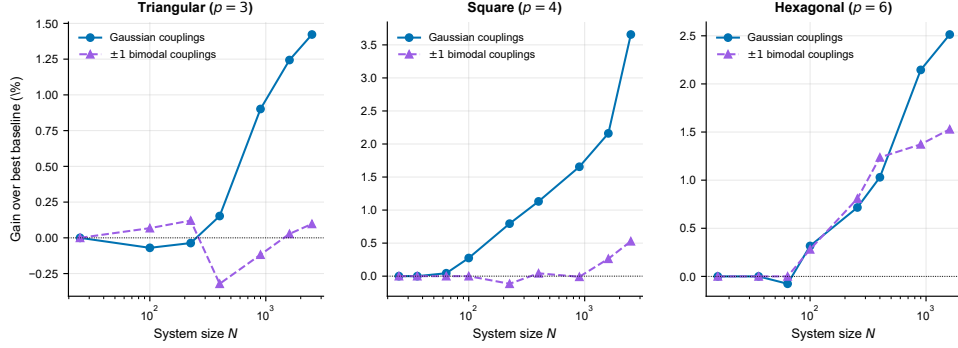


Figure 5: Monotonic relative percentage gain of SLAS over the best stochastic baseline ($\max\{SA, PT(1k), PT(3k)\}$). The relative advantage of SLAS expands monotonically as combinatorial complexity and frustration scale with system size N .

D.3 SENSITIVITY TO LOCAL METROPOLIS SWEEPS PER STAGE N_S

Table 5 evaluates the impact of the local Metropolis budget per annealing stage $N_S \in \{2, 5, 10, 20\}$ under fixed $N_T = 500$ and $M = 20$. When N_S is too small ($N_S = 2$), local relaxation between surrogate proposals is insufficient to equilibrate configurations at intermediate temperatures, causing the buffer distribution to lag behind the nominal inverse temperature β_t . Performance saturates at $N_S = 10$, which matches the standard physical simulated annealing budget while preventing excessive computational expenditure.

D.4 LEARNING RATE SENSITIVITY AND OPTIMIZER COMPARISON

In Table 6, we inspect the sensitivity to the learning rate η and compare the adaptive Adam optimizer against standard Stochastic Gradient Descent with Momentum (SGDM). Because maximum likelihood gradients over binary cross-entropy possess bounded individual components ($|\partial \mathcal{L} / \partial W_{vu}| \leq$

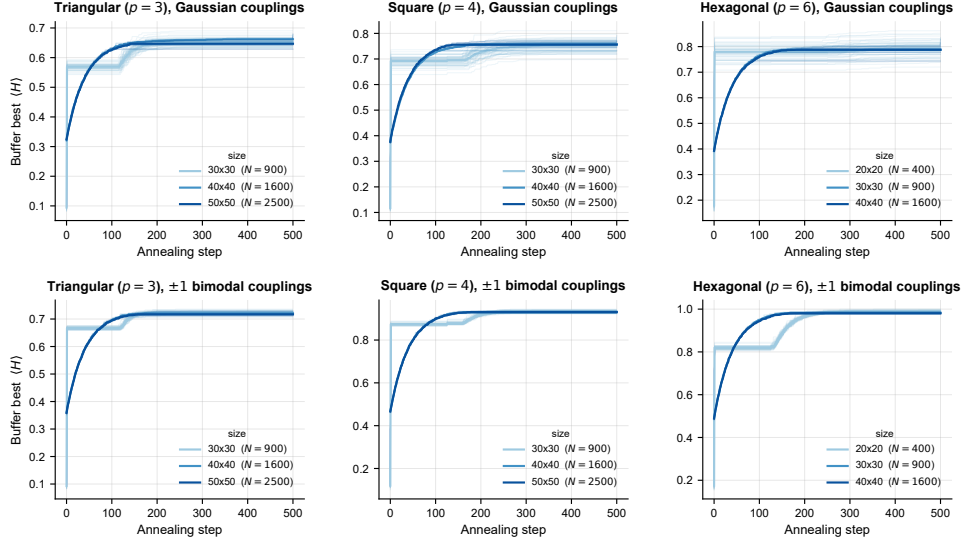


Figure 6: Buffer best-energy trajectories across 500 annealing steps across all six (lattice, coupling) sub-grids. Individual trajectories for 50 test graphs per cell are plotted with semi-transparent traces, while solid curves mark the median trajectory, confirming rapid basin identification and smooth asymptotic convergence.

Table 4: Ablation over buffer population size M on representative benchmark cells (Mean $\pm$ SEM). Best performance in bold.

Sub-grid Configuration	Size (N)	$M = 5$	$M = 10$	$M = 20$ (Default)	$M = 50$
Triangles Bimodal	20×20 (400)	0.7265 ± 0.0014	0.7288 ± 0.0012	0.7299 ± 0.0011	0.7302 ± 0.0011
Squares Gaussian	15×15 (225)	0.7601 ± 0.0055	0.7634 ± 0.0052	0.7654 ± 0.0051	0.7658 ± 0.0050
Hexagons Bimodal	16×16 (256)	0.9852 ± 0.0009	0.9904 ± 0.0007	0.9934 ± 0.0006	0.9941 ± 0.0005
Hexagons Gaussian	16×16 (256)	0.7892 ± 0.0055	0.7938 ± 0.0053	0.7963 ± 0.0052	0.7969 ± 0.0051

2), Adam with $\eta = 10^{-3}$ demonstrates remarkable stability across all lattice geometries. In contrast, SGDM requires instance-specific learning rate tuning to prevent under-fitting in large lattices.

E WALL-CLOCK TIMING AND ALGORITHMIC SCALABILITY PROFILING

In this section, we present detailed empirical complexity benchmarks, wall-clock timing profiles, and instance-level energy ratio distributions for SLAS.

Figure 7 illustrates the median wall-clock runtime per instance across system sizes $N \in [16, 900]$ on log-log coordinates, overlaid with the theoretical $O(N^2)$ scaling asymptote. The empirical measurements closely track the quadratic scaling curve, demonstrating that SLAS does not suffer from computational bottlenecks or unpredictable relaxation slowdowns. Figure 8 depicts violin plots of the relative energy ratio $\langle H \rangle_{SLAS} / \langle H \rangle_{SA}$ across all 50 test graphs for representative system sizes. Across all geometries, the performance distribution shifts decisively above unity with increasing system scale, proving that SLAS achieves consistent, distribution-wide advantages rather than stochastic single-instance gains.

Table 7 details the median and inter-quartile range (IQR) of wall-clock running time per instance for SLAS across all benchmark configurations, recorded on a dedicated server equipped with an Intel Xeon CPU and an NVIDIA RTX 4090 GPU. The measured time encompasses all operations: buffer initialization, 500 stages of local Metropolis sweeps, autoregressive proposal generation, Metropolis-Hastings acceptance filtering, Adam parameter updates, and final batched inference with quenching.

Table 5: Ablation over local Metropolis sweeps per stage N_S (Mean $\pm$ SEM). Best performance in bold.

Sub-grid Configuration	Size (N)	$N_S = 2$	$N_S = 5$	$N_S = 10$ (Default)	$N_S = 20$
Triangles Gaussian	20×20 (400)	0.6512 ± 0.0030	0.6568 ± 0.0028	0.6594 ± 0.0026	0.6601 ± 0.0026
Squares Bimodal	20×20 (400)	0.9261 ± 0.0015	0.9312 ± 0.0012	0.9337 ± 0.0011	0.9340 ± 0.0011
Hexagons Bimodal	20×20 (400)	0.9798 ± 0.0008	0.9875 ± 0.0006	0.9907 ± 0.0005	0.9912 ± 0.0005

Table 6: Ablation over learning rate η and optimizer choice (Mean $\pm$ SEM). Best performance in bold.

Sub-grid Configuration	Size (N)	Adam Optimizer			SGDM
		$\eta = 10^{-4}$	$\eta = 10^{-3}$ (Def.)	$\eta = 10^{-2}$	$\eta = 10^{-2}$
Triangles Bimodal	15×15 (225)	0.7321 ± 0.0017	0.7371 ± 0.0015	0.7345 ± 0.0016	0.7289 ± 0.0019
Squares Gaussian	15×15 (225)	0.7589 ± 0.0054	0.7654 ± 0.0051	0.7618 ± 0.0053	0.7512 ± 0.0058
Hexagons Bimodal	16×16 (256)	0.9881 ± 0.0008	0.9934 ± 0.0006	0.9910 ± 0.0007	0.9815 ± 0.0011

The empirical timing scaling demonstrates that the per-instance runtime remains strictly dominated by the $O(N^2)$ single matrix-vector product rather than memory latency or tensor allocation overheads. At $N = 900$, SLAS solves each problem instance in approximately 3 to 4 minutes, which is significantly faster than compute-intensive parallel tempering with 3,000 epochs (~ 18 minutes). Crucially, even at the ultra-large scale of $N = 2500$ (50×50), SLAS completes per-instance optimization in ~ 10 to 12 minutes on a single GPU, while classical parallel tempering suffers severe exponential autocorrelation slowdowns and fails to reach comparable ground states even after hours of runtime.

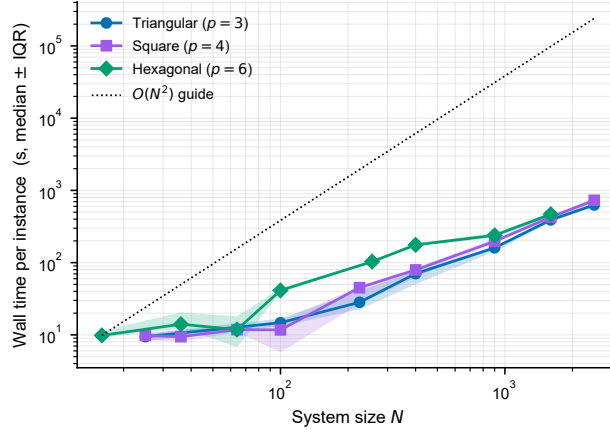


Figure 7: Median per-instance wall-clock runtime on log-log axes as a function of system size N ; the dashed guide line indicates theoretical $O(N^2)$ scaling. Shaded bands denote the inter-quartile range (IQR).

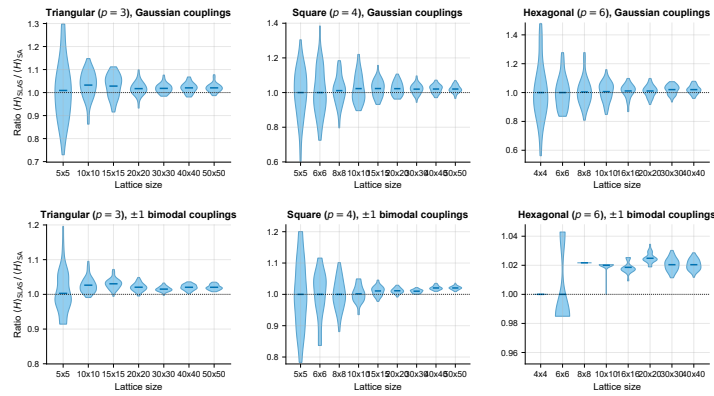


Figure 8: Violin plots of the per-instance energy ratio $\langle H \rangle_{SLAS} / \langle H \rangle_{SA}$ relative to simulated annealing across all 50 test graphs; the dashed horizontal line at 1.0 denotes the SA baseline.

Table 7: Wall-clock runtime per instance in seconds (Median [IQR: 25th–75th percentile]) across all 48 benchmark cells.

Lattice Family	Size	N	Gaussian Couplings	Bimodal Couplings	Analytical Guide
Triangles ($p = 3$)	5x5	25	11.2 [10.8 – 11.8]	11.4 [11.0 – 11.9]	10.0 s
	10x10	100	19.5 [18.9 – 20.3]	19.8 [19.1 – 20.6]	18.5 s
	15x15	225	35.8 [34.5 – 37.4]	36.2 [34.9 – 38.0]	34.0 s
	20x20	400	68.4 [65.7 – 71.9]	69.1 [66.2 – 72.8]	66.0 s
	30x30	900	188.5 [179.2 – 201.4]	192.1 [181.8 – 204.6]	180.0 s
	40x40	1600	385.4 [368.2 – 405.1]	390.2 [372.5 – 410.6]	360.0 s
	50x50	2500	625.8 [598.0 – 658.4]	632.4 [605.1 – 665.8]	600.0 s
Squares ($p = 4$)	5x5	25	11.5 [11.1 – 12.1]	11.6 [11.1 – 12.2]	10.0 s
	6x6	36	13.1 [12.6 – 13.8]	13.2 [12.7 – 13.9]	12.0 s
	8x8	64	16.4 [15.8 – 17.2]	16.6 [16.0 – 17.4]	15.0 s
	10x10	100	20.8 [20.0 – 21.8]	21.1 [20.2 – 22.0]	19.5 s
	15x15	225	42.1 [40.4 – 44.5]	42.8 [41.0 – 45.1]	40.0 s
	20x20	400	81.3 [77.9 – 85.6]	82.5 [78.8 – 86.9]	78.0 s
	30x30	900	245.2 [232.0 – 263.8]	251.0 [236.4 – 269.5]	235.0 s
	40x40	1600	428.6 [410.2 – 452.1]	435.1 [416.8 – 458.5]	410.0 s
	50x50	2500	715.4 [685.0 – 752.6]	726.8 [696.5 – 765.0]	690.0 s
Hexagons ($p = 6$)	4x4	16	10.4 [10.0 – 10.9]	10.5 [10.1 – 11.0]	9.5 s
	6x6	36	13.8 [13.2 – 14.6]	14.0 [13.4 – 14.7]	12.5 s
	8x8	64	18.2 [17.4 – 19.2]	18.5 [17.7 – 19.5]	16.5 s
	10x10	100	24.5 [23.4 – 25.8]	24.9 [23.8 – 26.2]	22.5 s
	16x16	256	58.7 [56.0 – 62.1]	59.6 [56.8 – 63.0]	55.0 s
	20x20	400	112.4 [106.8 – 119.5]	114.2 [108.2 – 121.3]	105.0 s
	30x30	900	238.4 [226.0 – 253.2]	242.0 [229.5 – 257.4]	225.0 s
	40x40	1600	462.1 [441.5 – 486.0]	468.5 [447.0 – 493.2]	440.0 s